Abstract

With the growing number of intelligent applications and services, it is critical to ensure the collected sensor data is of good quality, which has a direct effect on the efficacy of data-driven models. The scale and the wide spread of modern Internet of Things (IoT)-based sensor systems make it infeasible to inspect every sensor manually. It is also difficult to collect and label sufficient in situ data to train a good quality model that can correctly detect sensor data anomalies. Therefore, an unsupervised approach for detecting continuous sensor data anomalies is necessary. However, existing unsupervised anomaly detection techniques are vulnerable to concept drift and lack long-term reliability. The goal of this chapter is to examine existing unsupervised anomaly detection techniques for edge computing applications and investigate their potential to be integrated with concept drift detection to achieve long-term reliability for time series. This chapter concludes with a discussion of challenges and promising future research directions targeting long-term sensor data anomaly detection.